The Pseudo-Code for the FMA DFT Algorithm

Danyan Lin, Gang Wang, and K. C. Ho

Algorithm 1 FMA DFT Algorithm

Input:
  N: integer power of two DFT length;
  t: $\log_2 N$;
  $\mathcal{H}$: feasible set $\mathcal{N}$ or $\mathcal{P}$;
Calculate the reversed-bit order and collect them into a vector $\mathbf{r}_N$;
Determine the permutation matrix $\mathbf{P}_N$ from $\mathbf{r}_N$;
for $i = 1$ to t **do**
  Calculate $\mathbf{A}_i$ by (4), and set $\hat{\mathbf{A}}_t = \mathbf{A}_t$ and $\hat{\mathbf{A}}_{t-1} = \mathbf{A}_{t-1}$;
  Initialize $\hat{\mathbf{H}}_i = \mathbf{I}_N$;
  if $i < t-1$ **do**
    Map $\mathbf{A}_i$ to $\hat{\mathbf{A}}_i = f_{\mathcal{H}}(\mathbf{A}_i)$;
    if Using the feasible set $\mathcal{P}$
      Calculate the calibration matrix $\bar{\mathbf{H}}_i$ by (7);
      Find the value ϵ_i that is the minimum absolute value of the real and imaginary parts among all non-zero elements of $\bar{\mathbf{H}}_i$;
      Calculate the scaling factor α_i by $f_{\mathcal{H}}(1/\epsilon_i)$;
      Update $\hat{\mathbf{H}}_i$ by $\hat{\mathbf{H}}_i = f_{\mathcal{H}}(\alpha_i \bar{\mathbf{H}}_i)$;
    end if
  end if
end for
Calculate $\hat{\beta}$ by (10);
Output: The factorization matrices $\hat{\mathbf{H}}_i$ and $\hat{\mathbf{A}}_i$ and scaling factor $\hat{\beta}$.
